

Patent Infringement Analysis: US 11,573,697 B2 — Samsung "Unified Neural Network" Keystroke Prediction

Executive Summary

Bottom line: No third-party keyboard product is, on current public evidence, a literal, element-by-element infringer of US 11,573,697 B2 ("Key-Net"). The patent's combination of (i) a single unified neural network containing CNN + forward LSTM + backward LSTM + batch normalizer + softmax, (ii) explicit fusion of device-hardware factors (bezel size, edge-display presence) and sensor/posture features (tilt, finger size, finger count) into a feature vector, and (iii) numerical-normalization-by-unique-factor plus one-hot-encoding feature pre-processing is unusually narrow relative to how Gboard, SwiftKey, Apple QuickType, and Chinese IMEs are publicly described. The most plausible infringement candidates — under a doctrine-of-equivalents reading — are **Google Gboard** (LSTM/RNN spatial model + N-gram/NN-LM decoder, on-device touch normalization, posture/spatial personalization) and **Microsoft SwiftKey** (production GRU + experimental transformer with on-device inference and multi-feature context). Apple QuickType (bi-LSTM language model) and Chinese IMEs (Sogou/Baidu/iFlytek) match fewer elements. **No forward citations to US 11,573,697 have been identified** and no litigation or licensing activity tied to this patent has been found, suggesting the patent is currently a defensive/portfolio asset filed by Samsung R&D Institute-Bangalore (SRI-B).

TL;DR

- **Most exposed product: Google Gboard.** Google's own published architecture combines an on-device Neural Spatial Model (LSTM-class RNN) with touch-coordinate normalization, spatial personalization to user posture/finger placement, and probability-ranked key/word candidates — matching most-but-not-all claim 1 elements. **Severity: MEDIUM. Confidence: Medium.**
- **Second most exposed: Microsoft SwiftKey.** Microsoft documents a quantized on-device neural language model (GRU in production with 20K vocabulary and 512 hidden units; experimental transformer at ~6 MB after quantization per arXiv 2505.05648) plus on-device personalization. **Severity: LOW-MEDIUM. Confidence: Medium**, because public documentation does not show device-hardware features (bezel/edge-display) or sensor data feeding the model. (arXiv)
- **Likely non-infringers (as documented):** Apple QuickType (bi-LSTM language model only, no documented sensor/hardware feature fusion), Sogou, Baidu, iFlytek, Fleksy, Typewise — these match the unified-NN concept but lack the patent's distinctive device-hardware + sensor + posture feature aggregation. Samsung Keyboard Neural itself practices the patent; any Samsung-licensed OEM keyboard would have authorization. (Apple Machine Learning R...)

Patent Overview

Field	Value
Patent No.	US 11,573,697 B2
Title	"Methods and Systems for Predicting Keystrokes Using a Unified Neural Network"
Assignee	Samsung Electronics Co., Ltd.
Inventors	Barath Raj Kandur Raja; Ankur Agarwal; Bharath C; Harshavardhana; Ishan Vaid; Kranti Chalamalasetti; Mritunjai Chandra; Vibhav Agarwal (google)
US App No.	17/111,038
Filed	December 3, 2020 (google)
Granted	February 7, 2023 (google)
Pre-grant publication	US 2021/0173555 A1 (Jun. 10, 2021) (google)
Estimated expiration	December 9, 2040 (google)
Priority	Indian Provisional 201941022207 (Dec. 4, 2019); Indian complete (Jun. 4, 2020)
Foreign counterparts (confirmed)	India 201941022207
Foreign counterparts (NOT confirmed via Google Patents)	The task brief's WO2019/199942 and EP 20897022.8 could not be verified; WO2019/199942 was published in October 2019 — before the Dec. 4, 2019 Indian priority date — so it cannot be a family member.
Forward citations	Zero identified as of May 23, 2026
Litigation / licensing	None identified

Independent claims:

- Claim 1 (method):** obtain aggregated data from a plurality of factor types affecting typing behavior → extract features → form feature vectors (normalize numerical by unique factor; encode non-numerical) → predict key via unified neural network on probability. (uspto)
- Claim 7 (method):** receive keystroke → generate keystroke log → train unified NN to identify known features and learn unknown features by weight updates → predict using same normalization/encoding logic.
- Claim 12 (device):** memory + processor performing claim-1 operations.

Verbatim architectural recital from the specification:

"The unified neural network 301c includes at least one Convolutional Neural Network (CNN), at least one forward Long/Short-Term Memory Network (LSTM), at least one backward LSTM, at least one batch normalizer, and an activation layer." (google)

"In an example, consider that four feature vectors, touch positions, previous character position, sensor data, and temporal features, are supplied as input to the unified neural network 301c... The feature vectors, viz., touch position and the sensor data, have been obtained through normalization of their respective feature counterparts... The feature vectors, viz., previous character position and temporal features, have been obtained by encoding the respective feature counterparts." (google)

Claim Breakdown Matrix

Element	Type	Critical for Infringement?
(a) Obtain aggregated data from a plurality of factor types	Literal	YES — must show multiple factor types
(b) Factor types include any of: device characteristics, contextual device parameters, contextual user parameters, keyboard specifics, keystroke details, touch information	Dependent (claim 2/13)	One+ required
(c) Extract plurality of features	Literal	YES
(d) Determine numerical vs non-numerical per feature	Literal	YES (claim 1's wherein clause)
(e) Normalize numerical features by factor unique to the extracted feature	Literal	YES — this is a narrow limitation
(f) Encode (one-hot) non-numerical features	Literal	YES — narrow limitation
(g) Unified neural network predicts key via probability	Literal	YES
(h) NN comprises CNN + fwd LSTM + bwd LSTM + batch norm + activation/softmax	Spec/dependent	Literal claim 1 only requires "unified neural network"; this combination is recited in spec and several dependent claims
(i) Identify known features + learn unknown features (weight updates)	Claim 7	YES for claim 7
(j) Probability over multiple keys, highest selected	Claim 4/15	Common element

The combination most distinctive to the patent: numerical normalization "by a factor unique to the extracted feature" + non-numerical encoding + the unified network performing both touch decoding and language-context correction in one model.

Methodology of Investigation

1. Reviewed full patent specification and claims via USPTO PDF and Google Patents.
2. Surveyed published architecture documentation for each major keyboard product — prioritizing first-party engineering blogs (Google Research, Apple Machine Learning Research, Microsoft/SwiftKey arXiv) over secondary aggregators.
3. Searched for forward citations to US 11,573,697 via Google Patents and Justia; searched for related litigation across US/India/EPO/Korea via general legal news.
4. Mapped each product's documented architecture against the claim-breakdown matrix above.
5. Used a focused subagent to confirm forward-citation count, foreign-counterpart status, and pull verbatim spec passages.

Detailed Findings

1) Google Gboard (Google LLC) — MEDIUM severity, Medium confidence

Documentation:

- Google Research Blog, "The Machine Intelligence Behind Gboard" (May 2017): describes a Neural Spatial Model at character level + FST decoder for words; touch-point coordinate normalization across screen sizes; spatial features include location, speed, direction, key traversed. (Google Patents)
- Hard et al., "Federated Learning for Mobile Keyboard Prediction," arXiv 1811.03604 (2018): on-device CFIG (Coupled Input-Forget Gate, an LSTM variant) with **exactly 1.4 million parameters** (single-layer CFIG, 670 units, 10,000-word vocabulary, "more than two thirds of which are associated with the embedding matrix W"); shipped model is **exactly 1.4 MB after weight quantization**; training uses the **FederatedAveraging algorithm** (not plain federated SGD).
- Sivek & Riley, "Spatial Model Personalization in Gboard," Proc. ACM Hum.-Comput. Interact. 6 MHCI Article 202 (Sept. 2022; arXiv 2209.11311): personalizes a Gaussian spatial model by individual user offset means and learned covariances; explicit posture-aware modeling. (arxiv)
- Zhang et al., "Neural Search Space in Gboard Decoder," EMNLP-Industry 2024 / arXiv 2410.15575: replaces N-gram LM with NN-LM dynamically integrated into the C·L·G FST decoder. (arXiv)
- Google patent US 2016/0299685 A1 ("Neural network for keyboard input decoding"): explicitly describes LSTM RNN for keyboard input with spatial features (location, speed, direction, curvature) and bidirectional architecture. (Google Patents) (Google Patents)

Claim-by-claim mapping:

Claim 1 element	Gboard evidence	Match
Plurality of factor types	Touch position + previous characters + posture/spatial offset + device-screen size	✓
Extract features	Yes	✓
Numerical normalization by unique factor	Touch-point coordinates normalized across screen resolutions (Google Research 2017 blog)	✓ partial — "unique factor" not clearly per-feature in published docs
Encode non-numerical features	Character-level / token embeddings used	✓ functional
Unified NN with probability output	NSM (RNN/LSTM) + NN-LM (separate models combined via FST)	⚠ Two-stage architecture, not "unified" single NN as patent recites
CNN + fwd LSTM + bwd LSTM + batch norm + softmax	LSTM/bi-LSTM documented; CNN not standard in NSM; batch norm not publicly confirmed	⚠ Partial
Identify known + learn unknown features	Federated learning via FederatedAveraging updates model weights on device	✓

Reasoning: Gboard practices many elements but its public architecture decomposes the problem into a spatial model and a separate language model combined via FST, rather than a single "unified neural network" that performs both touch decoding and language-context correction (which is exactly what the patent's Key-Net does). Under literal infringement, the "unified" requirement is a meaningful distinction. Under doctrine of equivalents, the combined NSM+NN-LM system performs substantially the same function in substantially the same way. Critically, **Gboard does NOT publicly use device-hardware factors (bezel size, edge-display presence) nor on-device sensor data (tilt, accelerometer) for keystroke prediction** — these are dependent claim 2/9 factors and would limit damages to claim 1 if proven.

Severity: MEDIUM. **Confidence:** Medium.

2) Microsoft SwiftKey (Microsoft Corporation) — LOW-MEDIUM severity, Medium confidence

Documentation:

- Mike Butcher, "Swiftkey Releases Predictive Keyboard Built On A Neural Network," TechCrunch, October 8, 2015: SwiftKey Neural Alpha "is effectively a re-engineering of the Swiftly platform and an experimental project out of SwiftKey's 'skunkworks department' called Greenhouse." [TechCrunch](#) [TechCrunch](#)
- Tom's Hardware / TNW, 2016: SwiftKey rolls out neural network model running locally on-device. [TNW | Apps](#) [Tom's Hardware](#)
- Abouelenin et al., "Privacy-Preserving Transformers: SwiftKey's Differential Privacy Implementation," arXiv 2505.05648 (May 8, 2025): SwiftKey's production language model is a GRU within an engine called "Fluency"; the production GRU comparator has "20K vocabulary, 512 units in the hidden layer." An experimental transformer (scaled-down GPT-2-like) is described whose "total model size is around 6 MB after quantization"; integration via ONNX; DP fine-tuning on typing data. [arXiv + 6](#)

Claim mapping:

Element	SwiftKey evidence	Match
Aggregated data, plurality of factor types	Context tokens, previous typing, personalization	Partial — no documented sensor/hardware features
Numerical normalization by unique factor	Not documented publicly	Unknown
Non-numerical encoding	BPE sub-word tokenizer (~15K)	✓ functional
Unified NN predicting key by probability	GRU/transformer producing token probabilities	✓ for word/token, not character-key
CNN + fwd/bwd LSTM + batch norm + softmax	GRU (no CNN, no explicit bi-LSTM)	⚠ Different architecture
Identify known + learn unknown	On-device personalization + DP fine-tuning	✓

Reasoning: SwiftKey is the *closest archetype* of an on-device unified neural keyboard, but it is a language model (word/token prediction) rather than a spatial-decode model that takes touch features. There is no public documentation of CNN, batch normalization, or sensor/hardware feature fusion. Literal infringement is unlikely; doctrine-of-equivalents argument exists for the language-context-correction subset of claim functionality but not for the touch-decoding/sensor-aware subset.

Severity: LOW-MEDIUM. **Confidence:** Medium.

3) Apple QuickType (Apple Inc.) — LOW severity, Medium confidence

Documentation:

- Apple Machine Learning Research, "Can Global Semantic Context Improve Neural Language Models?": explicitly describes bi-LSTM architecture for QuickType keyboard semantic embeddings + neural language model. (Apple Machine Learning R...)
- AppleMagazine 2024–2025 and Apple Intelligence developer notes: on-device prediction on Neural Engine of A18/M4; integration with iOS 18+ AI text features. (AppleMagazine)

Claim mapping:

- Bi-LSTM matches "forward LSTM + backward LSTM" element.
- No CNN, no batch normalization documented.
- No documented use of device sensor data, bezel/edge-display features, or finger-size/finger-count.
- Numerical normalization by unique factor and one-hot encoding: not documented.
- Predicts on probability: yes.

Severity: LOW. **Confidence:** Medium. Apple practices the neural-LM concept but lacks the patent's distinctive multimodal-feature/sensor/hardware element aggregation.

4) Samsung Keyboard / Samsung Keyboard Neural Beta — Patentee's own product

The Samsung Keyboard Neural Beta app (APKMirror, com.samsung.android.honeyboard.neural) is explicitly marketed as using "a language model that has a neural network structure," with personalization and deep learning. As Samsung is the assignee, this is practice of the invention, not infringement. **Severity:** N/A. (APKMirror)

5) Chinese IMEs — Tencent Sogou, Baidu IME, iFlytek

- **Sogou (Tencent):** Major focus on Hunyuan large-model integration for AI search (KrASIA Nov 2024); historical core is statistical/NN word prediction. No public documentation of CNN+bi-LSTM+batch-norm architecture matching the patent. (KrASIA)
- **Baidu IME:** Baidu's deep-learning expertise (PaddlePaddle, Deep Speech RNN/LSTM) is well-documented for speech, but specific input-method architecture details are not public.
- **iFlytek:** Documented DNN/LSTM/UB-LSTM for **speech** input, not specifically for tap-keystroke prediction. CNN explicitly used to raise handwriting accuracy (Synced 2017).

(BaiduWiki) (Medium)

Severity: LOW. **Confidence:** Low (limited English-language documentation; would need Chinese-language patent search and IME version inspection). These products operate primarily in China, where US patent rights are non-enforceable; would need to confirm Korean or Indian patent counterparts to assert there.

6) Fleksy, Typewise, OpenBoard, AnySoftKeyboard

- **Fleksy:** "Fleksynext" neural deep-learning engine for context/sentiment-based app suggestions (TechCrunch 2019); core typing uses proprietary spatial error-correction

algorithms. No public CNN+bi-LSTM+batch-norm architecture documented. TechCrunch

- **Typewise:** Markets AI-driven typo reduction; on-device privacy-first design. No public architecture documentation matching patent. Dezeen
- **OpenBoard / AnySoftKeyboard:** Open-source; source-code inspection would be required. Generally do not implement neural prediction at the level required by the patent.

Severity: LOW (insufficient evidence to find infringement). **Confidence:** Low.

7) Xiaomi / OPPO / Vivo / OnePlus / Huawei / Honor / Realme keyboards

No first-party engineering documentation could be located describing on-device neural architecture for these OEMs' built-in keyboards. Many of these OEMs ship third-party keyboards (Sogou/Baidu) by default in China and Gboard/SwiftKey internationally. **Severity:** Insufficient evidence; **Confidence:** Low.

Forward Citation Analysis

As of May 23, 2026, **zero forward citations to US 11,573,697 B2 have been identified** on Google Patents or Justia. This is consistent with the patent being recently granted (Feb. 7, 2023) and operating in a narrow technical niche. The absence of citations also means there are no admissions of awareness by competitors via patent-prosecution practice. Independent confirmation via Lens.org or PatSnap is recommended.

Final Summary Table

Company / Product	Claims Implicated	Severity	Confidence	Evidence Count	Key Evidence
Google Gboard	Claim 1 (most elements; "unified" element disputable); claim 7 (federated learning maps to learning unknown features)	MEDIUM	Medium	5	Google Research blog 2017; arXiv 1811.03604 (federated CIFG: 1.4M params, 1.4MB quantized, FederatedAveraging); arXiv 2209.11311 (spatial personalization); arXiv 2410.15575 (neural search space); US20160299685
Microsoft SwiftKey	Claim 1 (language-context subset only); claim 7 (DP fine-tuning	LOW-MEDIUM	Medium	3	arXiv 2505.05648 (production GRU: 20K vocab, 512 hidden units; experimental transformer ~6MB quantized); TechCrunch Oct. 8, 2015

Company / Product	Claims Implicated	Severity	Confidence	Evidence Count	Key Evidence
	maps to weight updates)				(Butcher); Tom's Hardware 2016
Apple QuickType	Partial claim 1 elements (bi- LSTM only)	LOW	Medium	2	Apple ML Research bi- LSTM blog; AppleMagazine on-device coverage
Samsung Keyboard Neural	Practice by assignee	N/A	High	2	APKMirror; samsung.com support
Sogou / Baidu / iFlytek	Insufficient evidence	LOW	Low	2	KrASIA Nov 2024; Synced 2017
Fleksy / Typewise	Insufficient evidence	LOW	Low	2	TechCrunch 2019; Dezeen 2021
Xiaomi / OPPO / Vivo / OnePlus / Huawei / Honor / Realme	Insufficient evidence	Unknown	Low	0	—
OpenBoard / AnySoftKeyboard	Insufficient evidence	LOW	Low	0	—

Recommendations

Stage 1 — Validation (0-60 days):

1. Independently confirm zero forward citations via Lens.org, PatSnap, and USPTO Patent Center. Threshold to upgrade priority: ≥ 1 third-party forward citation.
2. Obtain the actual model files of Gboard and SwiftKey via APK decompilation/binary inspection to test whether the on-device model contains (i) a single graph combining CNN + LSTM + batch norm + softmax and (ii) per-feature normalization constants. If yes, severity for that product upgrades to HIGH.
3. Confirm or refute the Indian counterpart IN 201941022207 status (grant/pendency) via the Indian Patent Office InPASS database, since Gboard, SwiftKey, and Chinese IMEs are heavily used in India and would be the strongest enforcement venue alongside the US.

Stage 2 — Targeted infringement work product (60-180 days): 4. For Gboard: Prepare detailed claim chart for claim 1, comparing each clause against the Sivek & Riley spatial-personalization paper and the Hard et al. FederatedAveraging CIFG paper. Particular attention to the "factor

unique to the extracted feature" normalization language, which is narrower than the patent abstract suggests. 5. For SwiftKey: Subpoena or formal information request (if licensing discussions open) for the production model architecture diagram — public documentation only confirms the GRU (20K vocab, 512 hidden units) in production and the experimental ~6 MB transformer, leaving open whether CNN + batch normalization is used elsewhere in the stack. 6. Assess validity exposure: Search for pre-Dec. 2019 publications combining touch-spatial features with sensor/device-hardware features in a single neural net. Microsoft's Word Flow (2014–2016), Google's US20160299685, and academic CHI papers on posture-aware keyboards are candidates. If a strong §102/§103 reference exists, IPR risk should be modeled before any assertion.

Stage 3 — Decision points (180+ days): 7. If validation confirms a Gboard or SwiftKey model with all key-net architectural elements, consider a licensing approach rather than litigation, since (a) Samsung has cross-licensing relationships with both Google and Microsoft and (b) damages exposure across keyboards is bounded by per-unit royalties traditionally low in this space. 8. **Trigger to escalate to litigation:** any documented sensor-aware feature ingestion in a third-party keyboard (matches dependent claim 2/9), since this is the patent's most distinctive limitation and would significantly strengthen the literal-infringement argument.

Caveats and Limitations

1. **Public documentation gap.** First-party architecture descriptions for Gboard and SwiftKey are years old or research-flavored; production models may differ. This analysis can be definitively confirmed only with model-file inspection.
2. **"Unified neural network" construction risk.** A court could construe "unified" narrowly (a single end-to-end NN graph) or broadly (any composition of NN components serving keystroke prediction). Gboard's NSM+NN-LM combination is materially exposed under the broader construction and likely non-infringing under the narrower one. Samsung should expect heavy claim-construction litigation.
3. **Numerical-normalization-by-unique-factor.** This claim 1 wherein-clause is unusually specific. Many products do feature scaling but not necessarily per-feature unique factors; an accused infringer can argue this around with global standardization.
4. **Validity exposure.** The patent's combination of CNN + bi-LSTM + batch norm is conventional post-2017. Prior art combining touch features with bi-LSTM for keystroke decoding (US20160299685 Google; Microsoft Word Flow work) creates non-trivial §103 obviousness exposure, particularly in PTAB IPR if a competitor counters an assertion.
5. **Doctrine of equivalents reach.** Samsung's broadest reasonable reading would capture any neural keyboard that fuses multiple typing-context features and outputs key probabilities. Defendants would invoke prosecution history estoppel — Samsung's narrowing claim amendments (if any) during the 17/111,038 prosecution should be reviewed in the file history.
6. **Geographic enforceability.** US 11,573,697 only protects acts in the United States. Chinese IME usage in China is not infringing in the US; enforcement against Sogou/Baidu/iFlytek would require Chinese patent counterparts which have not been identified.

7. **Foreign-counterpart uncertainty.** The task's references to WO2019/199942 and EP 20897022.8 could not be confirmed as family members via Google Patents data; WO2019/199942's October 2019 publication actually predates the Dec. 4, 2019 Indian priority date, so it cannot be a family member. Only the Indian counterpart (IN 201941022207) is confirmed. The lead researcher should verify Espacenet/Global Dossier directly.
8. **No litigation or licensing baseline.** Because Samsung has not asserted this patent and no licensees are known, there is no royalty benchmark; any monetization plan must build a fresh damages case.